

Let's take a look at what the future might be, in the years to come. Expect some technologies which are offshoots of current display tech, or are built over current displays but offer a unique interface with which to interact with video content. Yet others are a completely new way to experience video.

The 2-way LCD

In 2007, Samsung showcased the first LCD panel which could operate two independent images on each side of its surface. The LCD product made use of Samsung's new double-gate, thin-film transistor (TFT) architecture. A TFT gate controls the liquid crystal alignment needed to reproduce on-screen images. Samsung's implementation had two gates that operate on each pixel instead of just one. Thus the screen on the front can display different images than the one on the back. Moreover, one of the screens was transmissive, allowing light to pass through, while the other was reflective in nature, throwing back light—this allowed the entire display to operate on just one backlight source. A 2-way LCD such as this could find its way onto the cell phones of tomorrow.

Something similar, yet completely different was showcased by Sharp towards the end of 2007. Sharp Corporation developed and showcased an LCD TV screen which can simultaneously display two independent images at the same time. Similar to the Samsung's but different in the detail that both the images are displayed on the same surface! The implantation is such that one particular image is seen visible when the LCD is viewed from a particular angle, while a completely different image is seen from a different viewing angle. Sharp employed a parallax barrier superimposed on an ordinary TFT LCD, to do the trick: the LCD sends the light from the backlight into right and left directions, making it possible to show different information and visual content on the same screen at the same time depending on the viewing angle. Such an LCD could find its way next to the driver's seat—offering one set of information to the driver, like a GPS map; and a completely different visual to the passenger, such as a movie.

Flexible displays

TVs built with Organic Light Emitting Diodes (OLED) don't seem far away. As suggested, OLED screens employ an

organic material as the light-emitting substrate. This makes an OLED screen self-lit, requiring no back light. This then goes a long way to endow an OLED with several benefits such as—better viewing angles, high picture contrast, better colour reproduction, deeper black levels, lower power consumption, thinner display profiles, and the ability to 'print' out displays. An OLED display can be rolled, or drawn into an elliptical or circular shape—imagine the entire ceiling covered with an OLED light source, or the surface of a table covered in one, or a wall showcasing an OLED-powered, wallpaper-thin television set—this is the future promised by OLED.

The drawback is that the lifetime of OLEDs is far shorter compared to LCD or PDP technologies. But OLED is an actively researched area and if the manufacturers were not monetarily occupied with churning out LCD panels, its drawbacks would have already been solved. The solutions can't be far though, and already HD sets built using OLED have been showcased by the likes of Samsung, Sony and LG. OLEDs currently find themselves on smaller devices such as cell phones, wristwatches, and music players. This technology has a good chance to make it as a replacement to LCD and PDP displays.

Another form of flexible display is the E-ink display. E-ink is already the technology of choice for e-book manufacturers—Amazon's Kindle, Sony's Book Reader, the iLiad, all employ e-ink as their displays. E-ink is amazing in that it does not require any power to keep it running, only to make screen refreshes, making it ideal for reading books. E-ink displays are constantly evolving—they are now bendable, rollable, and while currently monochromatic, colour is not too far away; even videos have been showcased running on E-ink displays.

Interactive Displays

We will take a more thorough look at this class of displays a little later in this cover story. An interactive display such as the Microsoft's Surface or TouchWall is likely to make its presence felt to us mere mortals in the coming years. These displays allow you to directly interact with what's on screen—generally via a touch screen or a multi-touch interface. Apart from the aforementioned solutions, you also have the Mitsubishi DiamondTouch which takes a unique approach: it is a tabular screen which consists of a series of sensors on its surface. But that's not the whole of it. It also consists of sensors placed near a user's body, such as a chair. Now when the user touches such a display, her body acts as a channel and connects the two sensors together. This allows the display to determine exact information on several fronts: which user touched it, which hand was used, and how many fingers.

The Phillips Entertaible is another table-display interface, but more aimed at gaming. It comprises a 32-inch horizontal LCD, which is enabled by touch screen-based multi-object position detection. The Entertaible offers electronic games which combine the features of computer gaming, such as dynamic playing fields and gaming levels, with the social interaction and tangible playing pieces, such as pawns and dice, of traditional board games.

Another unique interface is offered by the so-called



Different images seen from different viewing angles